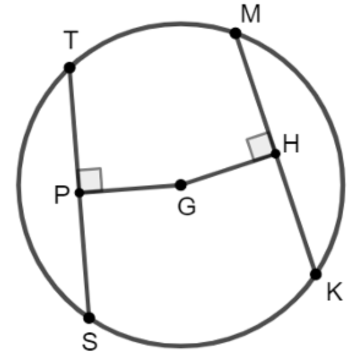
Geometry
Unit 12
Lesson 1 Practice

Name____________________
Date____________________
Period__________________

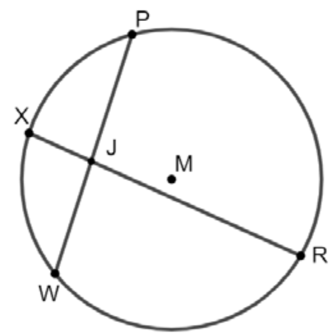
Circle G with points $T, M, K,$ and S on the circle are shown with chords TS and MK equidistant from the center. $TS = (9x + 6)$, $MK = (11x - 10)$, and $m\angle TPG = m\angle MHG = 90°$



1. Find the value of x.

2. What is the length of chord MK?

In Circle M, chords RX and PW intersect at point J. Points $R, W, X,$ and P lie on Circle M. $RJ = (8x + 3)$, $JX = 4$, $PJ = (4x - 2)$, and $JW = 12$.

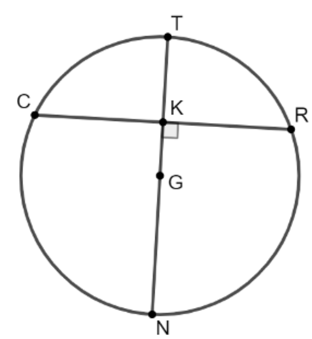


3. Solve for the value of x.

4. What is the length of line segment RJ?

5. Find the length of chord PW.

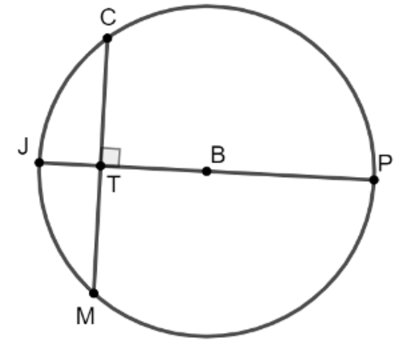
Circle G is shown where points $C, T, R,$ and N are on the circle. The diameter, NT, is a perpendicular bisector of chord CR. $CK = (5x + 6)$ and $KR = (7x - 2)$.



6. Calculate the value of x.

7. What is the length of chord CR?

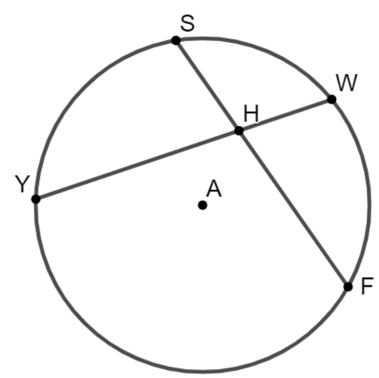
Chord CM is drawn in Circle B so that it intersects the diameter JP.



8. Given that $CM = 24$ and $TB = 5$, what is the length of BP?

Circle A is shown where points $Y, S, W,$ and F lie on the circle. Chords SF and YW intersect at point H. $YH = (4x + 2)$, $SH = 9$, $HW = 8$, and $HF = (4x)$. Mia and Nazir are solving for x and their work is shown below.

Mia's Work	**Nazir's Work**
$8(4x+2) = 9(4x)$ $32x + 2 = 36x$ $4x = 2$ $x = 0.5$	$8(4x+2) = 9(4x)$ $32x + 16 = 36x$ $4x = 16$ $x = 4$



9. Whose work do you agree with? Justify your answer.

In Circle H, points $P, L, M,$ and S lie on the circle. Chord PM intersects the diameter SL at point G and $m\angle HGP = 90°$.

10. If $PM = 32$ and $HS = 18$, what is the length of GH? Round to the nearest hundredth if necessary.